

# Preparation Enhances the Effectiveness of Positive Emotion Regulation

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The present research provides a new perspective on how preparation influences cognitive reappraisal to enhance positive emotions. Across two experiments, we examined whether the impact of preparation on positive emotion regulation (ER) is driven by the specific reappraisal tactic employed (self-focused reappraisal vs. situation-focused reappraisal) or by the overarching regulatory goal of enhancing positive emotions. In Experiment 1, 45 participants engaged in self-focused reappraisal, whereas in Experiment 2, 42 participants engaged in situation-focused reappraisal. An adapted ER task assessed emotional experiences under varying conditions, manipulating cues to either allow prior preparation (proactive ER) or occur without specific preparation (reactive ER). Results showed that reactive ER was consistently less effective than proactive ER in upregulating positive emotions, irrespective of the tactic employed. These findings support a *goal-driven account* over a *tactic-driven account*, highlighting the pivotal role of regulatory goals established during preparation.

**Keywords:** positive emotion regulation, preparation, cognitive reappraisal

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Positive emotions are discrete emotional states (e.g., joy, awe, gratitude) characterized by positive valence (Diener et al., 2020). In the course of human evolution, they have served a crucial adaptive function, acting as adaptive responses to survival-critical resources (e.g., food, social bonds) and growth-promoting opportunities (e.g., novelty, status elevation; Shiota et al., 2017). As Fredrickson (2001) posited, their role extends beyond momentary pleasure to drive behaviors that enhance long-term biological and social fitness. Therefore, understanding how to regulate positive emotions is vital.

We have introduced a temporal dimension to the understanding of emotion regulation (ER; J. Wang, Li, et al., 2024; J. Wang et al., 2025) demonstrating that preparation—mental readiness preceding the downregulation of negative emotions—plays a critical role. While these findings underscore the importance of preparation in negative ER, it remains unknown how preparation influences the upregulation of positive emotions, which facilitate adaptation to environmental demands and promote well-being. In this study, we address a foundational

question: whether preparation impairs or enhances the effectiveness of positive ER.

## Two Primary Reappraisal Tactics in ER

ER, as defined by Gross (1998), is the process by which individuals manage emotions—choosing which emotions to feel, when to experience them, and how to express them. A major ER strategy is cognitive reappraisal, which modifies the emotional impact by reinterpreting the meanings of stimuli or situations (Brudner et al., 2018). Cognitive reappraisal comprises two primary tactics: self-focused reappraisal, which involves adopting new perspectives to alter one's psychological distance toward the stimuli and modify their emotional impact, and situation-focused reappraisal, which entails reframing the context or meaning of the stimulus (Powers & LaBar, 2019; Willroth & Hilimire, 2016). Both tactics can regulate negative and positive emotions (Ochsner & Gross, 2005). However, due to the

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unique characteristics of positive emotions, their regulation emphasizes enhancing immediate experiences rather than diminishing them, as with negative emotions (Bryant & Veroff, 2017). These differing regulatory goals underscore the need to examine how self-focused and situation-focused reappraisals uniquely operate in positive ER contexts.

The majority of studies comparing self-focused and situation-focused reappraisals have focused on the regulation of negative emotions. Several studies indicate that situation-focused reappraisal is more effective than self-focused reappraisal at downregulating negative emotions (Hermann et al., 2021; Qi et al., 2020; Willroth & Hilimire, 2016), whereas meta-analyses reveal no significant behavioral differences in regulatory effects between the two tactics (Denny et al., 2023; Webb et al., 2012). From a neuroscientific perspective, the two tactics recruit distinct neural pathways: Situation-focused reappraisal mainly engages the left ventrolateral prefrontal cortex (PFC), a key region for cognitive control (Denny et al., 2023; Messina et al., 2015), whereas self-focused reappraisal involves extensive regions of the default mode and frontoparietal networks, which support psychological distance representation and self-projection (Powers et al., 2020; Powers & LaBar, 2019).

### How Preparation Shapes Regulation in Cognitive Reappraisal: Tactic-Driven Versus Goal-Driven Account

Although cognitive reappraisal is frequently studied as a unified construct in empirical research, theoretical models posit that it is a multicomponent process (Ford et al., 2017). It begins with deciding to use this strategy and continues through its successful implementation (Gross, 2015; Kreibitz et al., 2023). Accordingly, the ER process can be broadly divided into two stages: preparation and regulation. Previous studies on negative ER have shown that self-focused reappraisal paired with preparation enabled earlier and more effective ER, while preparation for situation-focused reappraisal may delay its onset and impair its regulatory effect (J. Wang, Li, et al., 2024; J. Wang et al., 2025). The previous pattern of results can be summarized as a *tactic-driven account*: The impact of preparation on regulation is determined by the tactic rather than the regulatory goal. Drawing on the broaden-and-build theory, we speculate that engagement in the ER task activates adaptive attentional narrowing when processing negative emotional stimuli (Fredrickson, 2004). In turn, this attentional narrowing facilitates the concentrated allocation of cognitive resources, enabling individuals to tailor their preparations to the unique characteristics of the tactic before regulation.

Additionally, research on positive ER via cognitive reappraisal remains underdeveloped. One study demonstrated that positive ER allows individuals to alter emotional experiences, facial expressions, and physiological responses (e.g., heart rate and respiration) according to their regulatory goals (Giuliani et al., 2008). During positive ER, cognitive reappraisal engages the lateral PFC and dorsomedial PFC, with prefrontal-subcortical couplings associated with successful regulation (Kim & Hamann, 2007; Li et al., 2018). However, prior research has neither directly compared self-focused and situation-focused reappraisal in the context of positive emotions nor examined the impact of preparation on these tactics. This gap leaves us without a clear understanding of how preparation influences the effectiveness of cognitive reappraisal for positive emotions.

In contrast to negative ER, which focuses on reducing or eliminating negative emotional experiences, positive ER often seeks to amplify

positive emotional responses, thereby buffering against adverse events and promoting psychological well-being (Fredrickson & Levenson, 1998; Gross, 2015; Sokołowski et al., 2022). The broaden-and-build theory posits that individuals experiencing positive emotions tend to adopt a global cognitive configuration, indicative of a broadened thought repertoire (Fredrickson, 2004). Given the intrinsic drives underlying human behavior and their alignment with task demands, we postulate that in positive ER, the preparation stage is oriented not merely toward the efficient execution of tactics but rather toward achieving the overarching goal of upregulating positive emotions (Higgins, 2011). Therefore, in positive ER, the relationship between preparation and regulation may be described by a *goal-driven account*: The influence of preparation on regulation is determined by the regulatory goal rather than by the tactic. Notably, when upregulating positive emotions, cognitive reappraisal—compared with other strategies—activated regions such as the dorsolateral PFC, supramarginal gyrus, angular gyrus, and precuneus, which are key for goal-directed behavior and information assimilation (Tsujimoto et al., 2022). Thus, the regulatory goal may shape the relationship between preparation and regulation more than specific tactics.

### The Present Study

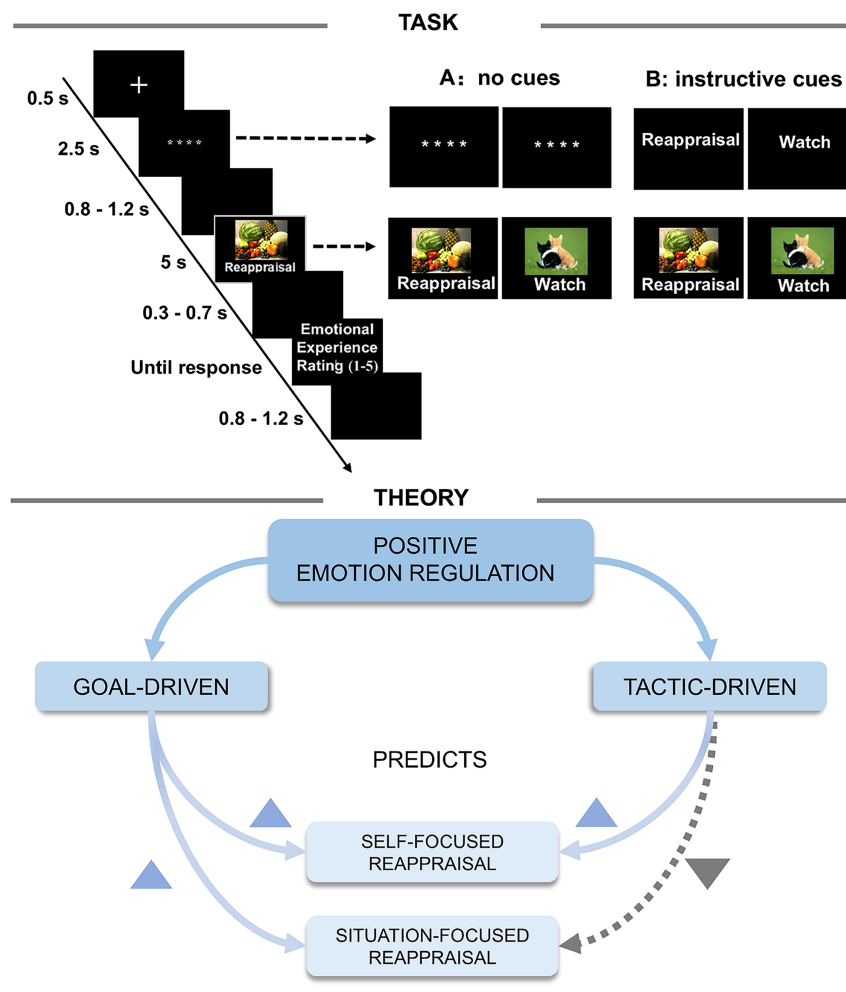
We propose two accounts of how preparation influences regulation, each offering different predictions for the tactics used in positive ER. Across two experiments, participants completed the adapted ER task (J. Wang et al., 2022), using two tactics (self-focused and situation-focused reappraisal) to upregulate positive emotions. The task was divided into two ER modes: reactive ER, where participants received no prior cues and engaged in regulation without targeted preparation, and proactive ER, where participants were given cues before the task, allowing for preparation. We aim to determine whether the influence of preparation on regulation is tactic-driven or goal-driven across modes. If our findings support the *tactic-driven account*, preparation will enhance regulation during self-focused reappraisal but impair regulation during situation-focused reappraisal. Consequently, when using self-focused reappraisal to upregulate positive emotions, proactive ER will outperform reactive ER; conversely, when employing situation-focused reappraisal, reactive ER is expected to be more effective. Alternatively, if our research supports the *goal-driven account*, preparation will enhance regulation for both tactics, making proactive ER superior regardless of the reappraisal tactic employed (see Figure 1).

## Method

### Overview of Studies 1 and 2

In Experiment 1, we tested the regulatory effect on participants to upregulate positive emotional experiences through self-focused reappraisal in both reactive and proactive ER. In Experiment 2, we used the same design to examine the regulatory effect on participants through situation-focused reappraisal. We collected ratings of emotional experiences during an adapted ER task designed to contrast reactive and proactive regulation dynamics. Each trial began with a fixation cross followed by a mode-specific cue: In reactive ER trials, an asterisk cue appeared before the emotional stimuli, requiring participants to prepare and implement the reappraisal strategy concurrently with the picture presentation; in proactive ER trials, an instructional cue

**Figure 1**  
Trial Procedure and Theoretical Predictions



*Note.* Upper panel: Trial sequence of Experiments 1 and 2. In reactive emotion regulation, participants received asterisks as prompts, while in proactive emotion regulation, they were given instructive cues. When emotional stimuli were presented, the instructive cues appeared below the images. Lower panel: Theoretical predictions. According to the *goal-driven account*, preparation should enhance the regulatory effect for both tactics. According to the *tactic-driven account*, preparation would facilitate regulation through self-focused reappraisal but impair it through situation-focused reappraisal. See the online article for the color version of this figure.

appeared in advance, allowing participants to prepare before picture onset. During picture display, participants either naturally observed the emotional content or applied cognitive reappraisal tactics that had been trained before the experiment. Finally, they rated the emotional intensity on a 5-point scale (1 = *not at all*, 5 = *extremely positive*) using the numeric keys. A key question was whether preparation would improve or impair the regulatory effect. Because both experiments employed the same ER task but different reappraisal focuses, we report their common procedures together.

## Participants

The sample comprised 45 participants (25 women and 20 men; age range = 18–22 years;  $M_{\text{age}} = 20.18$ ,  $SD = 1.37$ ) in Experiment

1 and 42 participants (23 women and 19 men; age range = 18–24 years;  $M_{\text{age}} = 20.55$ ,  $SD = 1.60$ ) in Experiment 2. All participants were native Chinese-speaking college students recruited through advertisements. Inclusion criteria were consistent across both experiments: Participants were right-handed, had normal or corrected-to-normal vision, and had no history of psychiatric or neurological disorders. No data exclusions occurred in either experiment. All participants provided written informed consent and received monetary compensation for their participation. This research adhered to the ethical standards set by the local Human Ethics Committee for Human Research (H22094).

We conducted an a priori power analysis using G\*Power 3 (Faul et al., 2007) to estimate the sample size required for our primary hypothesis, which focused on comparing the regulatory effects of

two modes of ER via a two-sided paired-sample  $t$  test in both experiments. Input parameters were set to  $\alpha = .05$  and  $1 - \beta = 0.8$ . Because no prior studies have directly examined the influence of preparation on positive ER, we selected a medium effect size ( $d = 0.5$ ) based on Cohen's conventions to estimate the required sample size for our behavioral study (Cohen, 1988, 1992). This analysis indicated a required total sample size of 34.

## Procedure

Upon receiving the task instructions, participants completed a practice session to familiarize themselves with the application of various types of instructions. To assess their understanding of the instructions, they had to report on how they implemented each trial type. Following this, they proceeded with the ER task.

Each trial commenced with a 500-ms presentation of a fixation cross, followed by the cue for 2,500 ms. For reactive ER, the asterisk served as the cue; for proactive ER, "watch" or "reappraisal" served as the instructional cues. After a variable duration (800–1,200 ms), a picture accompanied by an instructional cue was presented for 5,000 ms, during which participants executed the required task. In the watching condition, participants were directed to let natural thoughts and emotions emerge while viewing the pictures. In the reappraisal condition, for the first experiment, participants were instructed to increase their subjective closeness to the depicted events by vividly imagining themselves engaging in the actions portrayed and sharing in the positive emotions of the people depicted; in the second experiment, participants were instructed to reinterpret each picture to enhance its positive aspects by amplifying the positive content (see Supplemental Material S1). After a variable duration of 300–700 ms, participants rated the intensity of their positive emotional experiences with no time limit. The next trial started after a variable intertrial interval of 800–1,200 ms.

The task comprised reactive and proactive ER blocks. We employed a block design with a counterbalanced order: Some participants completed the reactive ER block first, while others began with the proactive ER block. Each ER block contained 108 trials, split equally into three sets of 36 trials. Each trial set consisted of three conditions: watch-neutral, watch-positive, and reappraisal-positive. Each condition included 36 trials. Trials within each set were presented in random order. We averaged the emotional experience ratings reported by each participant under each condition, using this mean as the statistical indicator for that condition. Upon completing the task, participants were asked to indicate whether they adhered to the provided regulation tactic and to evaluate its effectiveness in upregulating their positive emotions (see Supplemental Material S2).

## Stimuli

We selected 72 neutral pictures ( $M_{\text{valence}} = 5.25$ ,  $M_{\text{arousal}} = 4.45$ ) and 144 positive pictures ( $M_{\text{valence}} = 6.92$ ,  $M_{\text{arousal}} = 6.03$ ) from the Chinese Affective Picture System to use as emotional stimuli in two experiments (Bai et al., 2005). The stimuli, measuring  $374 \times 280$  pixels, were displayed with E-Prime (Schneider et al., 2002) on a 19-in. monitor (display resolution:  $1,024 \times 768$  pixels; refresh rate: 60 Hz). We randomly allocated all the pictures to two sets designed for different ER modes, ensuring both stimulus sets were comparable in valence ( $t_s < 1.19$ ,  $p_s > .23$ ) and arousal ( $t_s < 1.07$ ,  $p_s > .29$ ). There

was a significant difference in valence ( $t_s > 22.37$ ,  $p_s < .001$ ) and arousal ( $t_s > 15.43$ ,  $p_s < .001$ ) between positive and neutral pictures. An extra set of four positive and two neutral pictures was included in a practice block, with each picture presented just once in the ER task. Participants used the number keys to rate their emotional experiences.

## Data Analysis

To compare regulatory effects between reactive and proactive ER modes, we performed paired-samples  $t$  tests. Regulatory effects were defined as the difference in emotional experiences between the reappraisal-positive and watch-positive conditions. This approach has been widely employed in related studies (e.g., Giuliani et al., 2008; Langeslag & Surti, 2017; Y. Wang, Shanguan, et al., 2024) to assess the extent of ER. Effect sizes were calculated using Cohen's  $d$ .

We conducted a  $2 \times 3$  repeated-measures analysis of variance (ANOVA) to examine emotional experiences with factors "mode" (reactive ER vs. proactive ER) and "condition" (watch-neutral vs. watch-positive vs. reappraisal-positive). For factors with more than two levels, we assessed sphericity violations and applied the Greenhouse-Geisser correction when necessary. Significant interactions were further analyzed using post hoc tests to evaluate pairwise differences, with the Bonferroni correction applied to control the familywise error rate. Effect sizes for ANOVA results are reported as partial eta-squared ( $\eta_p^2$ ).

Additionally, we conducted exploratory analyses to investigate potential factors contributing to differences in regulatory effects. Pearson correlations were computed between emotional experiences in the watch-neutral condition and regulatory effects across ER modes. To assess the precision of these correlation estimates, we calculated 95% confidence intervals (CIs) for correlation coefficients using Fisher's  $z$ -transformation.

## Transparency and Openness

In line with transparency and openness, we detail our sample size determination, data inclusions, manipulations, and measures used in this study. The study was not preregistered regarding its design and analysis. Data analyses were conducted using SPSS Version 26 (IBM) and R Version 4.4.1. All analyzed data and code are available from the corresponding author upon reasonable request. We utilized Copilot and Grammarly to check for spelling and grammatical errors.

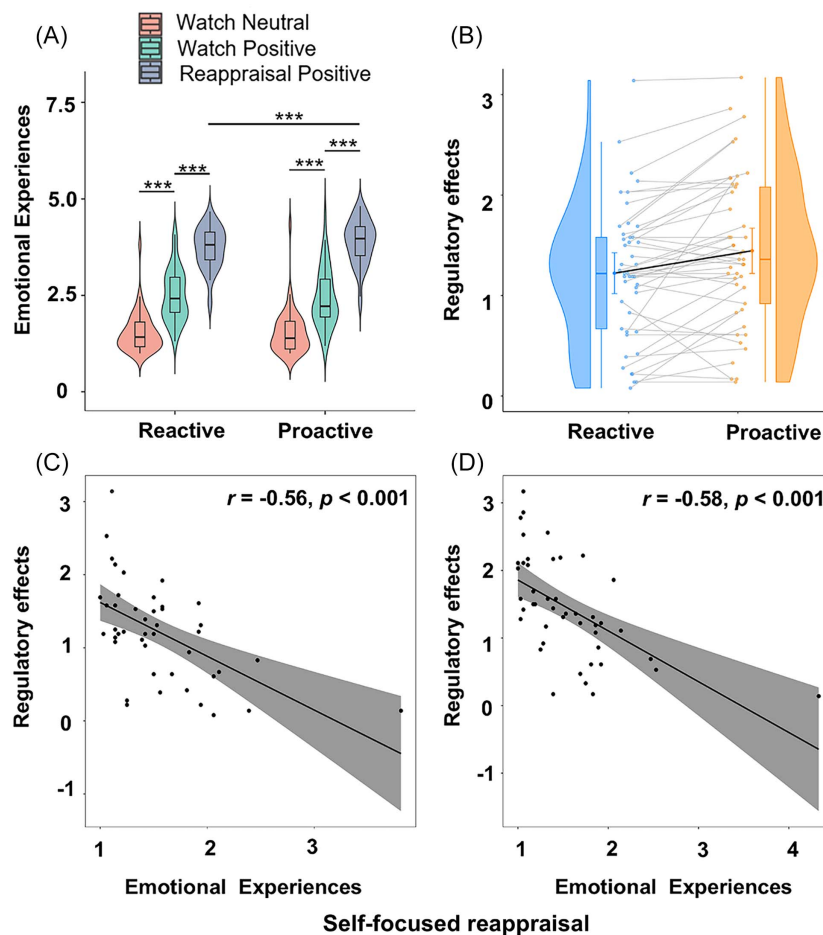
## Results

### Effects of Reappraisal

We conducted a paired-samples  $t$  test to investigate the difference in regulatory effects between reactive and proactive ER in Experiment 1. The analysis compared emotional experience ratings in the reappraisal-positive and watch-positive conditions. Result revealed that the regulatory effect in the proactive ER ( $M = 1.45$ ,  $SE = 0.11$ ) was better than that of the reactive ER ( $M = 1.22$ ,  $SE = 0.10$ ),  $t(44) = 3.85$ ,  $p < .001$ ,  $d = 0.57$  (see Figure 2B).

The descriptive results of Experiment 1 are presented in Figure 2A. The two-way ANOVA results are detailed in Table 1. Notably, the interaction between mode and condition was significant. Follow-up analyses of this significant interaction revealed that, in the reappraisal-positive condition, the difference between

**Figure 2**  
Main Results of Experiment 1



*Note.* (A) Results of condition comparisons across both modes. (B) Results of regulatory effect comparisons across both modes. The gray lines indicate individual data, while the black lines show the mean. (C, D) Correlations between emotional experiences in the watch-neutral condition and regulatory effects in both modes. Higher levels of emotional experience were associated with lower regulatory effects in both reactive and proactive emotion regulation. Shaded areas represent the 95% confidence interval of the regression line. See the online article for the color version of this figure.

\*\*\*  $p < .001$ .

proactive and reactive modes was significant ( $p < .001$ ). Specifically, the emotional experiences in the proactive mode ( $M = 3.88, SE = 0.086$ ) were more positive than those in the reactive mode ( $M = 3.73, SE = 0.080$ ), indicating that proactive mode upregulation was more effective. Furthermore, emotional experiences in the reappraisal-positive condition were more positive than those in the watch-positive condition in both reactive ( $M = 2.50, SE = 0.10; p < .001$ ) and proactive modes ( $M = 2.43, SE = 0.11; p < .001$ ). Additionally, emotional experiences in the watch-positive condition were more positive than those in the watch-neutral condition in both reactive ( $M = 1.54, SE = 0.076; p < .001$ ) and proactive modes ( $M = 1.55, SE = 0.087; p < .001$ ).

We also conducted a paired-samples  $t$  test to investigate the difference in regulatory effects between reactive and proactive ER in Experiment 2. Result revealed that the regulatory effect in the

proactive ER ( $M = 1.06, SE = 0.10$ ) was greater than that of the reactive ER ( $M = 0.88, SE = 0.091$ ),  $t(41) = 3.73, p = .001, d = 0.58$  (see Figure 3B).

The descriptive results of Experiment 2 are presented in Figure 3A. The two-way ANOVA results are detailed in Table 2. Notably, the interaction between mode and condition was significant. Follow-up analyses revealed that in the watch-neutral condition, reactive and proactive modes differed significantly ( $p = .01$ ). Specifically, emotional experience ratings were higher in the reactive mode ( $M = 1.68, SE = 0.11$ ) than in the proactive mode ( $M = 1.59, SE = 0.11$ ). Similarly, in the watch-positive condition, reactive and proactive modes differed significantly ( $p = .003$ ), with higher ratings in the reactive mode ( $M = 2.90, SE = 0.11$ ) than in the proactive mode ( $M = 2.75, SE = 0.13$ ). Additionally, emotional experiences in the watch-positive condition were more positive than those in the watch-

**Table 1**

*Experiment 1: Inferential Statistical Values for the Two-Way Repeated-Measures Analysis of Variance With Factors Mode (Reactive, Proactive) and Condition (Watch-Neutral, Watch-Positive, Reappraisal-Positive)*

| Effect                  | Emotional experience |          |          |            |
|-------------------------|----------------------|----------|----------|------------|
|                         | <i>df</i>            | <i>F</i> | <i>p</i> | $\eta_p^2$ |
| Mode                    | 1, 44                | 1.29     | .26      | 0.028      |
| Condition               | 2, 88                | 322.86   | <.001    | 0.88       |
| Mode $\times$ Condition | 2, 88                | 9.12     | <.001    | 0.17       |

*Note.* *df* = degrees of freedom.

neutral condition in both modes ( $ps < .001$ ). However, emotional experiences in the watch-positive condition were less positive than those in the reappraisal-positive condition in both reactive ( $M = 3.78$ ,  $SE = 0.11$ ;  $p < .001$ ) and proactive modes ( $M = 3.81$ ,  $SE = 0.12$ ;  $p < .001$ ).

### Relationship Between Performance in the Watch-Neutral Condition and Regulatory Effects

Given the significant main effect of condition in Experiment 1, we conducted correlation analyses to examine the relationship between the regulatory effect and emotional experience ratings in the watch-neutral condition. In reactive ER, we found a significant negative correlation,  $r = -0.56$ ,  $p < .001$ , 95% CI  $[-0.73, -0.31]$ ; in proactive ER, we also found a significant negative correlation,  $r = -0.58$ ,  $p < .001$ , 95% CI  $[-0.75, -0.35]$ . These results indicated that greater regulatory effects were associated with lower emotional experiences in the watch-neutral condition (see Figure 2C and 2D).

We conducted the same analyses in Experiment 2. In reactive ER, we found a significant negative correlation,  $r = -0.34$ ,  $p = .029$ , 95% CI  $[-0.58, -0.037]$ ; in proactive ER, we also found a significant negative correlation,  $r = -0.38$ ,  $p = .013$ , 95% CI  $[-0.61, -0.088]$ . These results again indicated that greater regulatory effects were associated with lower emotional experiences in the watch-neutral condition (see Figure 3C and 3D).

### Discussion

This study explored the influence of preparation on cognitive reappraisal in upregulating the positive emotions, using self-focused reappraisal in Experiment 1 and situation-focused reappraisal in Experiment 2. An adapted ER task allowed us to assess emotional experiences under different cue types designed to either enable prior preparation (proactive ER) or trigger immediate regulation without specific preparation (reactive ER). Across two experiments, proactive ER was more effective than reactive ER in upregulating positive emotions, regardless of the specific reappraisal tactic used. These findings suggest that the regulatory goal, rather than the specific tactic, plays a dominant role in shaping the effectiveness of positive ER.

We contrasted two theoretical accounts of positive ER. The *tactic-driven account* predicts a tactic-dependent pattern: Preparation should enhance the regulatory effect for self-focused reappraisal but impair it for situation-focused reappraisal. In contrast, the *goal-driven account* predicts that preparation consistently enhances regulatory effects across

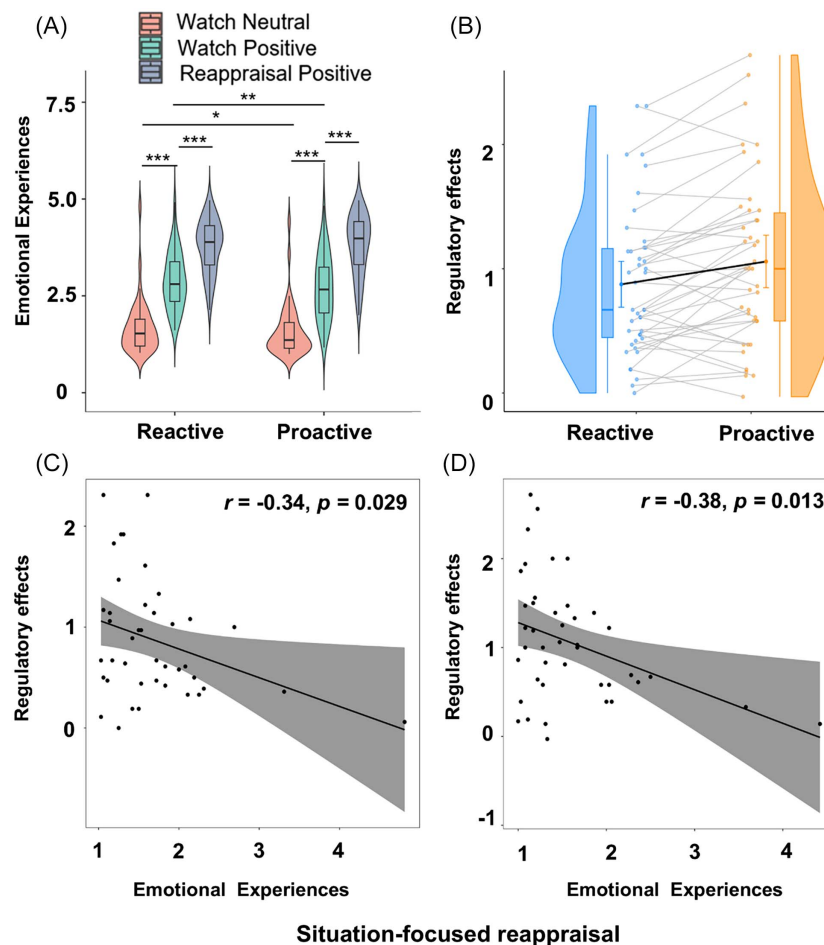
both tactics. Our results support the *goal-driven account*, demonstrating that preparation (proactive ER) enhanced regulatory effectiveness regardless of tactic. However, we acknowledge the established strength of the *tactic-driven account* in explaining negative ER (J. Wang, Li, et al., 2024; J. Wang et al., 2025). This divergence suggests unique characteristics distinguish the mechanisms underlying positive ER.

Specifically, participants employed self-focused reappraisal to regulate their positive emotions in Experiment 1. Compared to reactive ER, proactive ER significantly increased positive emotions, suggesting that preparation directly enhanced the regulatory effect. This finding aligns with a previous study, in which participants used self-focused reappraisal to downregulate the negative emotions (J. Wang, Li, et al., 2024). With this tactic, participants are asked to engage in self-projection and adopt a new perspective to reinterpret their relationship with the depicted events (Buckner & Carroll, 2007; Willroth & Hilimire, 2016). The establishment of upregulation goals prompted individuals to activate semantic knowledge and various memory associations, developing a self-referential perspective (Powers & LaBar, 2019; J. Wang, Li, et al., 2024). During regulation, participants adopted an alternative perspective, and the binding between the self-referential perspective and emotional stimuli enhanced the depth of emotional stimulus processing (Sui & Humphreys, 2015), allowing individuals to upregulate positive emotions effectively while preserving sufficient cognitive resources to fully experience these emotions.

In Experiment 2, proactive ER was also more effective than reactive ER in upregulating positive emotions via situation-focused reappraisal. First, compared to reactive ER, proactive ER provided participants with more time to activate hedonic motives, enabling them to establish regulatory goals and mobilize attentional resources during the preparation stage. Second, the effectiveness of regulation may depend on the motives individuals pursue (Tamir, 2016). During regulation, participants could utilize sufficient cognitive resources to reinterpret the event in a more flexible and positive manner to enhance pleasure, leading to better outcomes than those achieved through reactive ER. We observed significant improvements under proactive ER conditions, highlighting the importance of preparation. These findings indicate that the upregulation goal established during preparation is more critical than the specific tactic employed. Interestingly, under watch-positive conditions, emotional experiences were less positive in proactive ER than in reactive ER. In such positive states, individuals naturally strive to maintain their current mood, resulting in greater allocation of attention and effort to feeling good (Kashdan et al., 2015). However, under watch-positive conditions, participants still needed to process the pictures to understand their content and evoke natural emotions. Participants engaged in proactive ER may develop a plan about cognitive resource allocation before the presentation of images, facilitating more effective resource utilization during image processing. This process, however, may reduce the cognitive resources available for positive emotional experiences, ultimately leading to their reported diminishment.

Therefore, we propose that the effectiveness of positive ER was best explained by a *goal-driven account*. Given individuals' natural tendency to pursue positive emotional experiences, the preparation stage in both experiments provides sufficient time to transform this motivation into clear upregulation goals. The *goal-driven account* allocated more attentional resources during preparation, and during

**Figure 3**  
Main Results of Experiment 2



*Note.* (A) Results of condition comparisons across both modes. (B) Results of regulatory effect comparisons across both modes. The gray lines represent individual participant data, while the black lines represent the mean. (C, D) Correlations between emotional experiences in the watch-neutral condition and regulatory effects in both modes. Higher levels of emotional experience were associated with lower regulatory effects in both reactive and proactive emotion regulation. Shaded areas indicate the 95% confidence interval of the regression line. See the online article for the color version of this figure.

\*  $p < .05$ . \*\*  $p < .01$ . \*\*\*  $p < .001$ .

subsequent regulation, these resources were deeply engaged with emotional stimuli, thereby enhancing regulation effectiveness.

Both experiments revealed significant negative correlations between emotional experience ratings in the watch-neutral condition and regulatory effects in both ER modes. Lower ratings in the neutral condition may reflect a less positive emotional state. This could, in turn, heighten the desire for positive emotional experiences, as individuals generally strive to maintain or enhance pleasant emotions while minimizing unpleasant ones (Gross et al., 2006; Kashdan et al., 2015; Riediger et al., 2009). In reappraisal conditions, participants had the opportunity to upregulate positive emotions, which likely strengthened their motives to flexibly engage cognitive resources for reappraisal and produced more effective regulatory outcomes (Tamir, 2016). A contrast effect often occurs in self-judgments of emotional states (Manstead et al., 1983). After

experiencing heightened positive emotions, subsequent exposure to neutral stimuli may make the relative lack of positive emotions more salient. Consequently, this engenders a stronger motive to upregulate positive emotions via cognitive reappraisal. These findings imply that the strength of an individual's motive to regulate emotions may influence the effectiveness of regulation.

In everyday life, moments of joy can often feel fleeting, and many of us wonder how to prolong and deepen our positive emotions. This study highlights the crucial role of regulatory goal during the preparation stage in regulating positive emotions. Regulatory goals drive individuals' preparation for ER, whether through self-focused reappraisal through perspective shifting or through reinterpreting a situation, resulting in greater feelings of happiness. These findings offer insights for personal emotional management and suggest new approaches for psychological interventions. By encouraging individuals

**Table 2**

*Experiment 2: Inferential Statistical Values for the Two-Way Repeated-Measures Analysis of Variance With Factors Mode (Reactive, Proactive) and Condition (Watch-Neutral, Watch-Positive, Reappraisal-Positive)*

| Effect                  | df    | Emotional experience |       |            |
|-------------------------|-------|----------------------|-------|------------|
|                         |       | F                    | p     | $\eta_p^2$ |
| Mode                    | 1, 41 | 3.98                 | .053  | 0.088      |
| Condition               | 2, 82 | 282.46               | <.001 | 0.87       |
| Mode $\times$ Condition | 2, 82 | 7.10                 | .001  | 0.15       |

Note. df = degrees of freedom.

to prepare before engaging with positive experiences, we can enhance their ability to maintain well-being in today's fast-paced world. This research provides a practical tool for cultivating calm and lasting happiness amid modern life's challenges.

### Limitations and Future Directions

While this study advances our understanding of positive ER, three primary limitations warrant consideration. First, the restricted age range of our sample (predominantly young adults) constrains the generalizability of our findings. Previous research suggests that age-related declines in cognitive functions may affect the ability to regulate emotions through cognitive reappraisal (Kunzmann et al., 2022). Future studies should examine how age moderates the observed effects. Second, we relied on self-report measures to assess emotional experiences. While self-reports are valuable tools, they may not fully capture the complexity of emotional responses. Future research should include physiological measures (e.g., heart rate, electromyography, skin conductance) to provide a more comprehensive evaluation of emotional experiences. Third, this study focused on behavioral assessments. Future research should incorporate event-related potentials or functional magnetic resonance imaging to elucidate the neural mechanisms underlying positive ER.

### Conclusion

This study demonstrates that cognitive reappraisal is more effective in upregulating positive emotions when participants are provided with preparation time. It also demonstrates that this enhancement from preparation occurs regardless of the specific reappraisal tactic used, supporting a *goal-driven account*. Importantly, the findings suggest the critical role of establishing regulatory goals during the preparation stage of positive ER. These results underscore the importance of allocating sufficient preparation time before implementing cognitive reappraisal tactics. They further suggest that future interventions should aim to enhance individuals' motivation to regulate their emotions. This research offers novel insights into proactive preparation for positive ER for the development of comprehensive ER interventions and clinical treatment plans.

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